Supporting Information for

Making the Most of Little Time: A Probabilistic Approach to Marine Heatwave Detection

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Contents of this file

Figures S0.1 to 0.2: Additional Figures for Site South WA
Figures S1.1 to 1.9: Figures for Site Japan
Figures S2.1 to 2.9: Figures for Site North West Shelf, Australia
Figures S3.1 to 3.9: Figures for Site UK

Introduction

This Supporting Information provides additional diagnostic and model output figures for the South Wa, Japan, North West Shelf, and UK sites. These figures are equivalent to those presented in the main text for South WA and are included here to demonstrate the consistency of model behaviour and sampling diagnostics across all study locations.

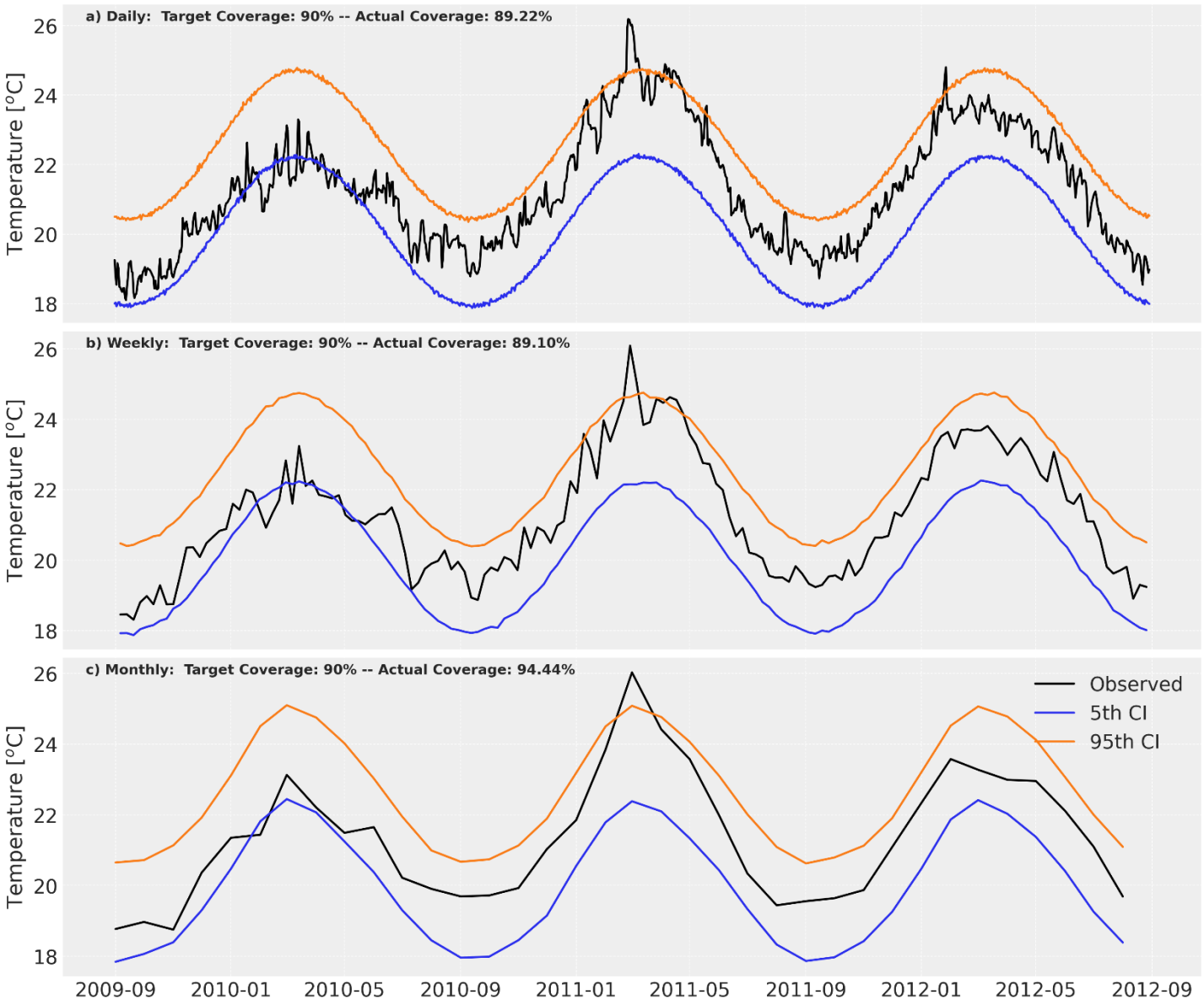


Figure S0.1. Posterior predictive calibration for a 3-year observational subset at the South WA site. The plots display the actual observed temperature against the model's 5th and 95th predictive credible intervals (representing a 90% target coverage) for a) daily, b) weekly, and c) monthly temporal resolutions.

To evaluate the statistical reliability of the Bayesian parameter inference framework, we conducted a posterior predictive calibration test. A well-calibrated probabilistic model should generate predictive uncertainty bounds that capture the true observed data at a rate matching the specified probability level. If the model is overconfident, too few observations will fall within the bounds; if it is underconfident, too many will.

For this analysis, we defined a target probability coverage of 90%. For every individual time step in the time series, the 5th and 95th percentiles were extracted from the simulated posterior predictive samples (the *likelihood* distribution) to construct a 90% predictive credible interval. We then calculated the empirical coverage, defined as the exact percentage of actual temperature observations that fell within these upper and lower bounds.

To visualise this fit, Figure S0.1 plots the 90% credible intervals against the raw observed temperature data for a representative 3-year subset at the South WA site, demonstrating the model's boundary tracking across daily, weekly, and monthly resolutions.

To systematically evaluate how data resolution, subset length, and prior specification influence this calibration, we computed the Coverage Error. This metric is defined as the empirical coverage minus the target 90% coverage (e.g., an empirical coverage of 91.5% yields an error of +1.5%). Figure S0.2 compiles the coverage error across subset windows, comparing both weak and strong priors against a strict ±2% tolerance corridor.

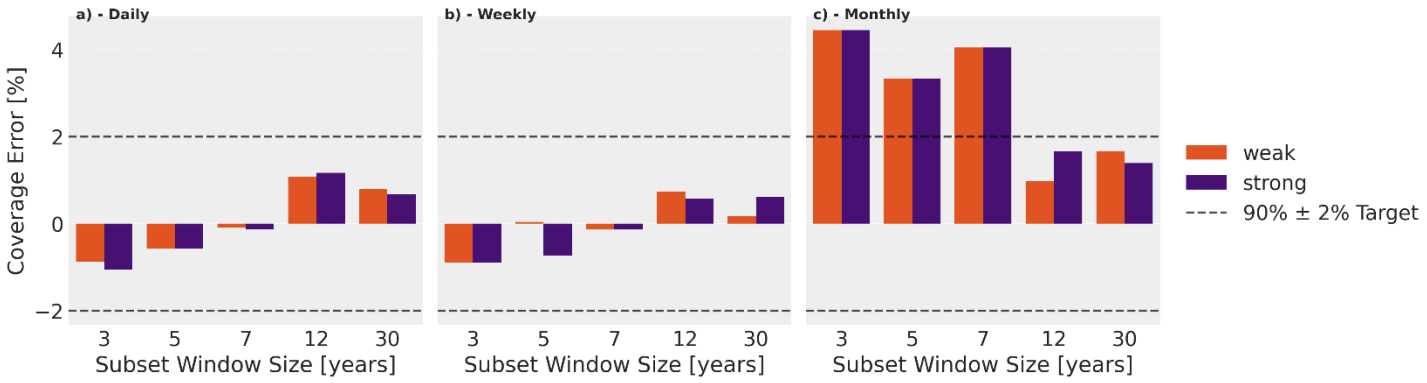


Figure S0.2. Calibration coverage error of the posterior predictive distribution across varying subset window sizes at the South WA site. The coverage error is calculated as the model's empirical coverage minus the target coverage of 90%. Results are compared between weak and strong prior specifications (as described in Section 3.4.2) for a) daily, b) weekly, and c) monthly temporal resolutions. Dashed lines indicate a ±2% error tolerance corridor.

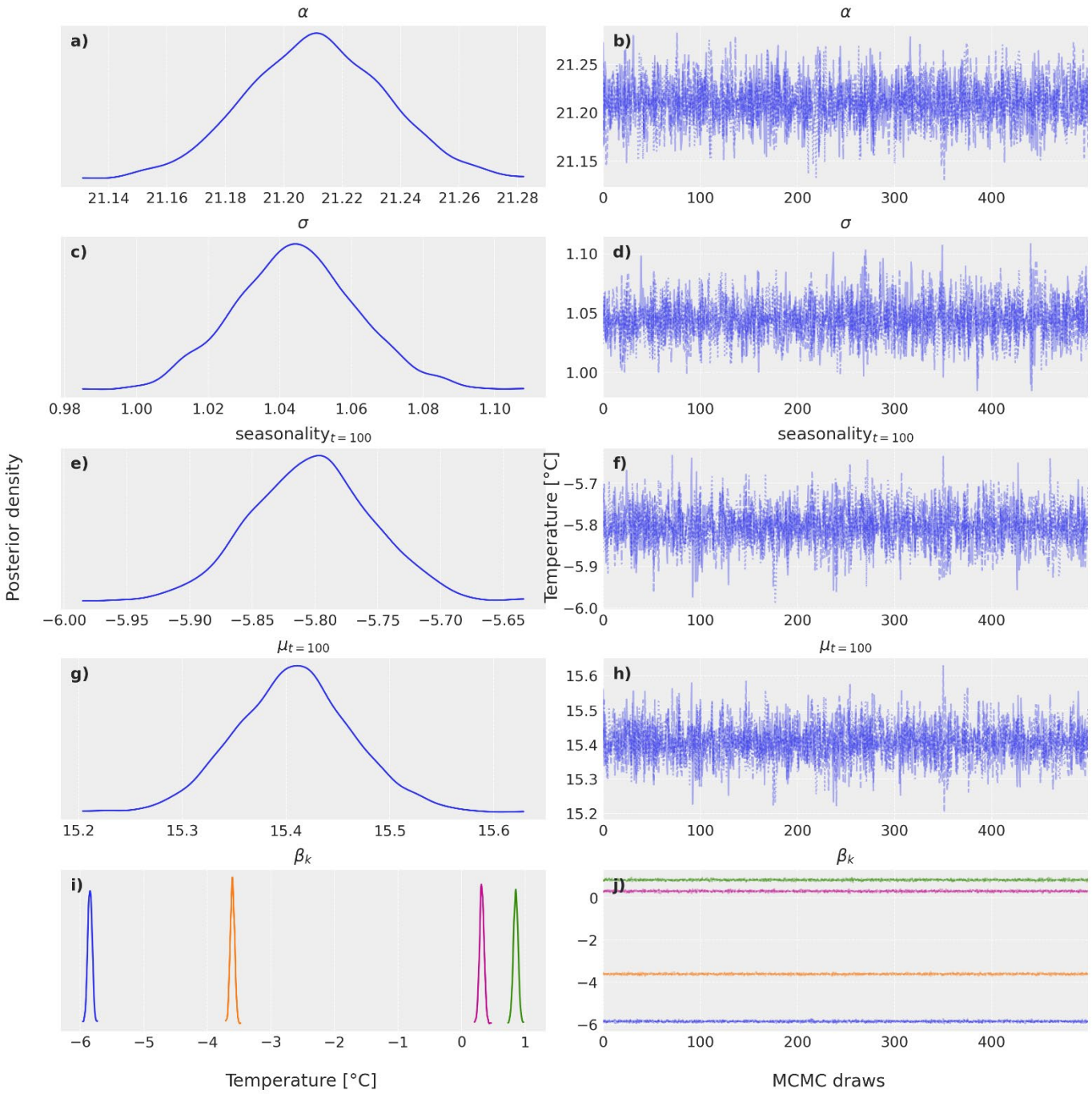


Figure S1.1. Same as Figure 4 but for site Japan**.**

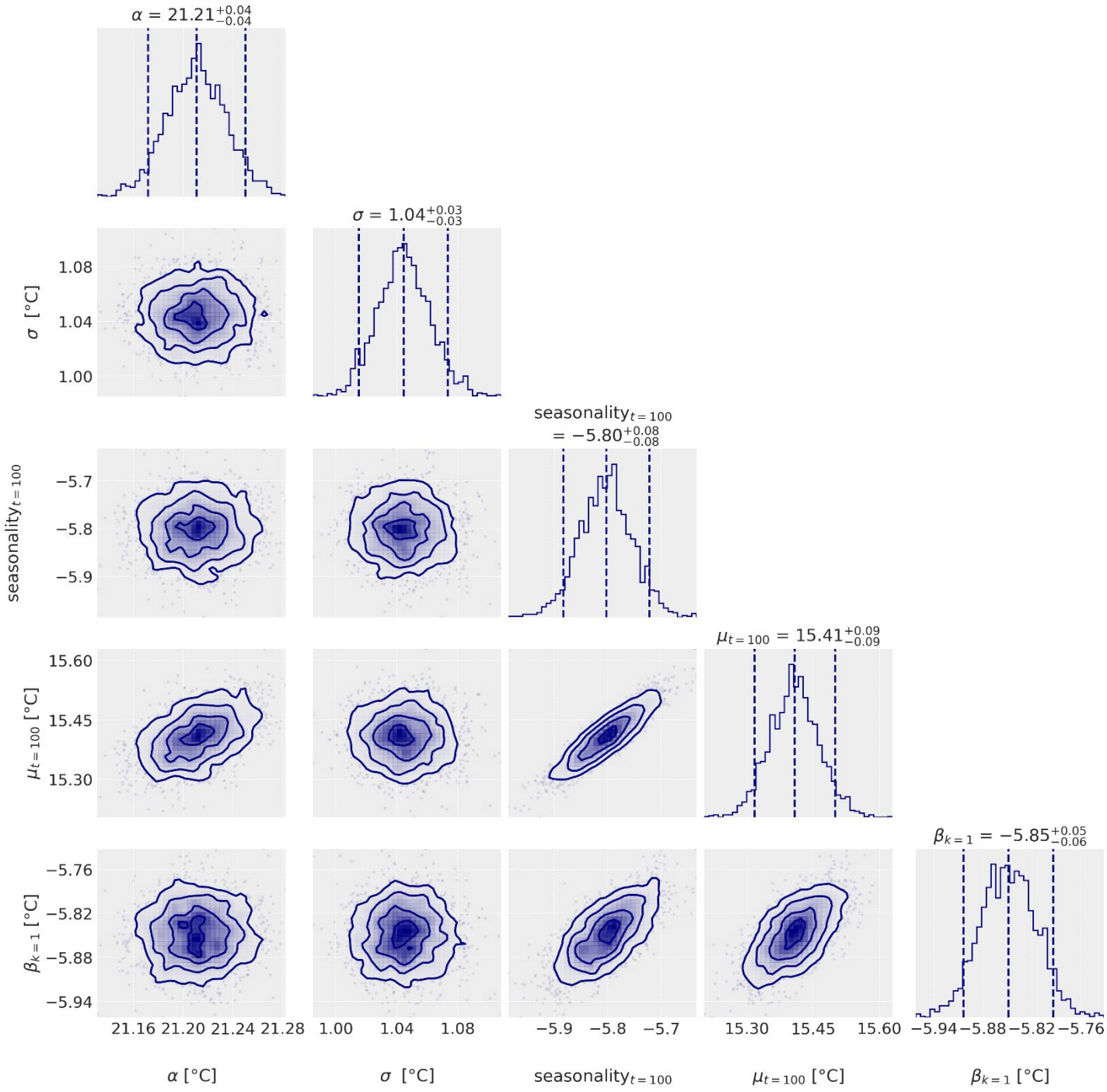


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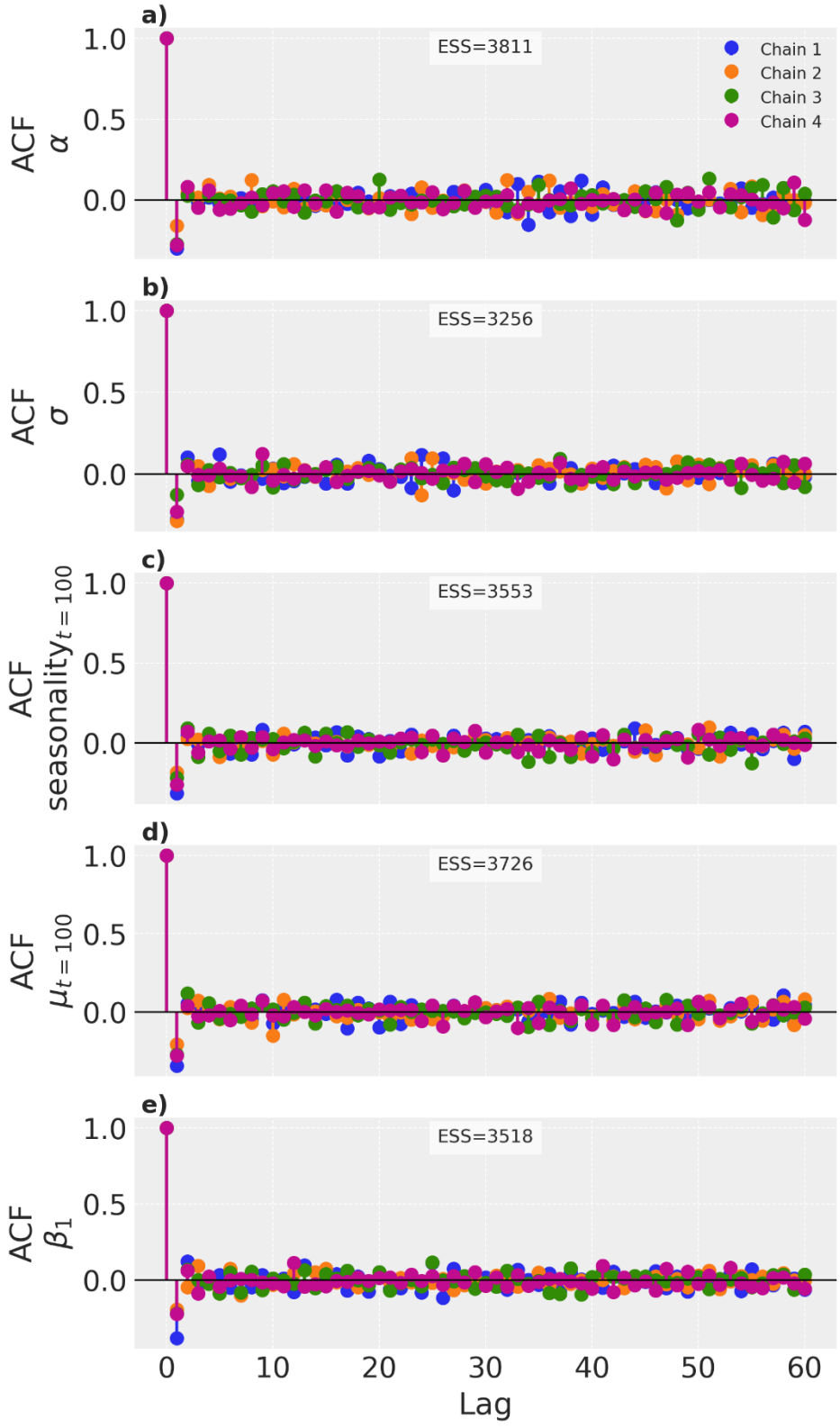


Figure S1.3. Same as Figure 6 but for site Japan.

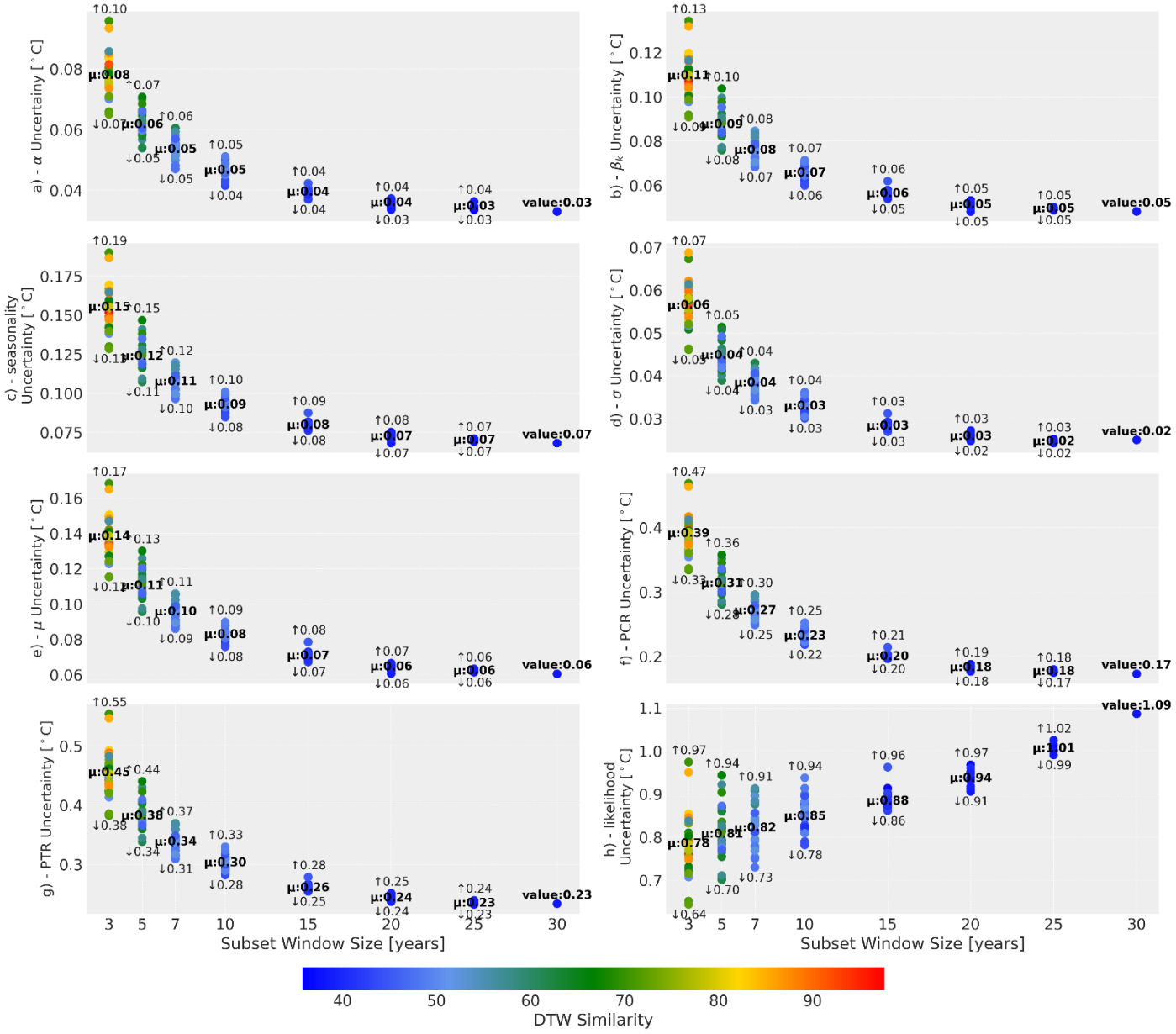


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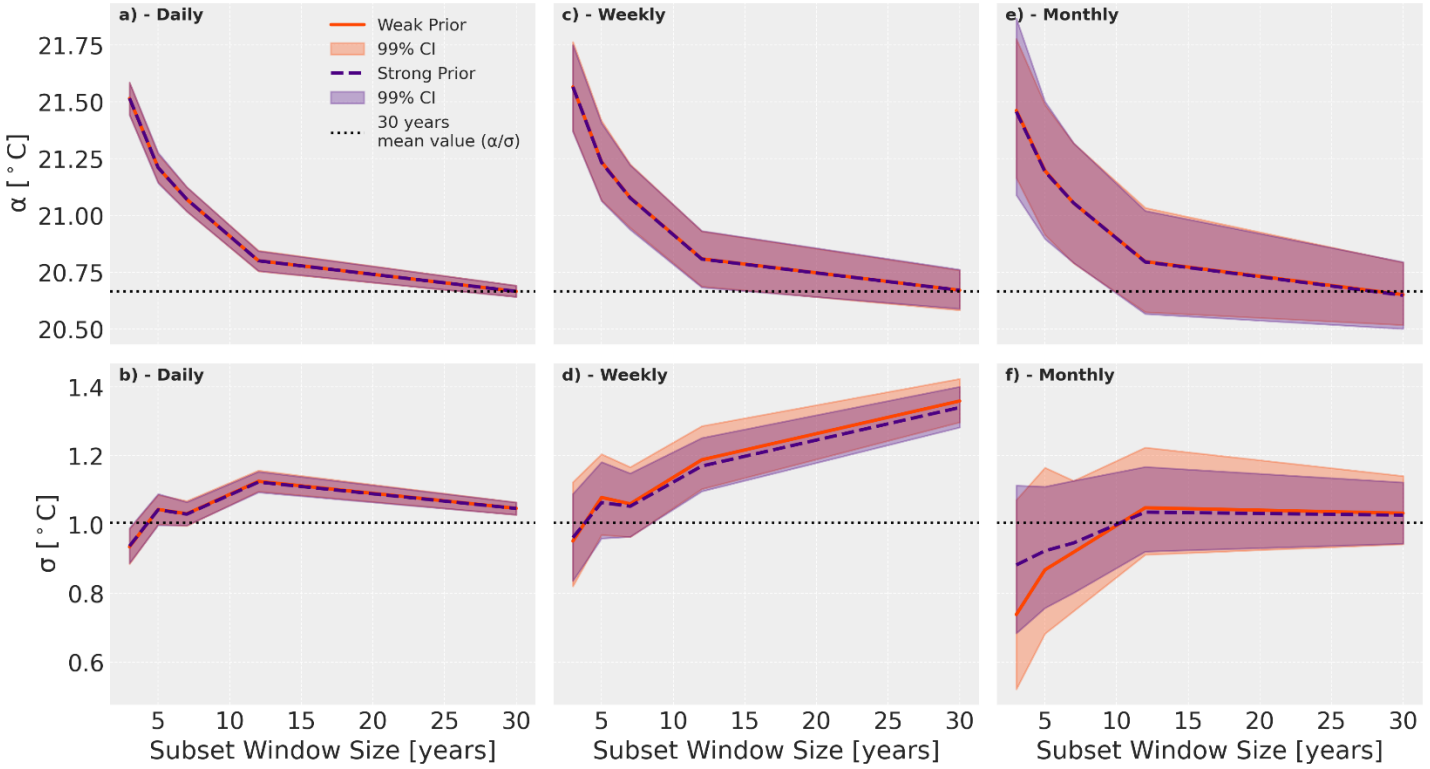


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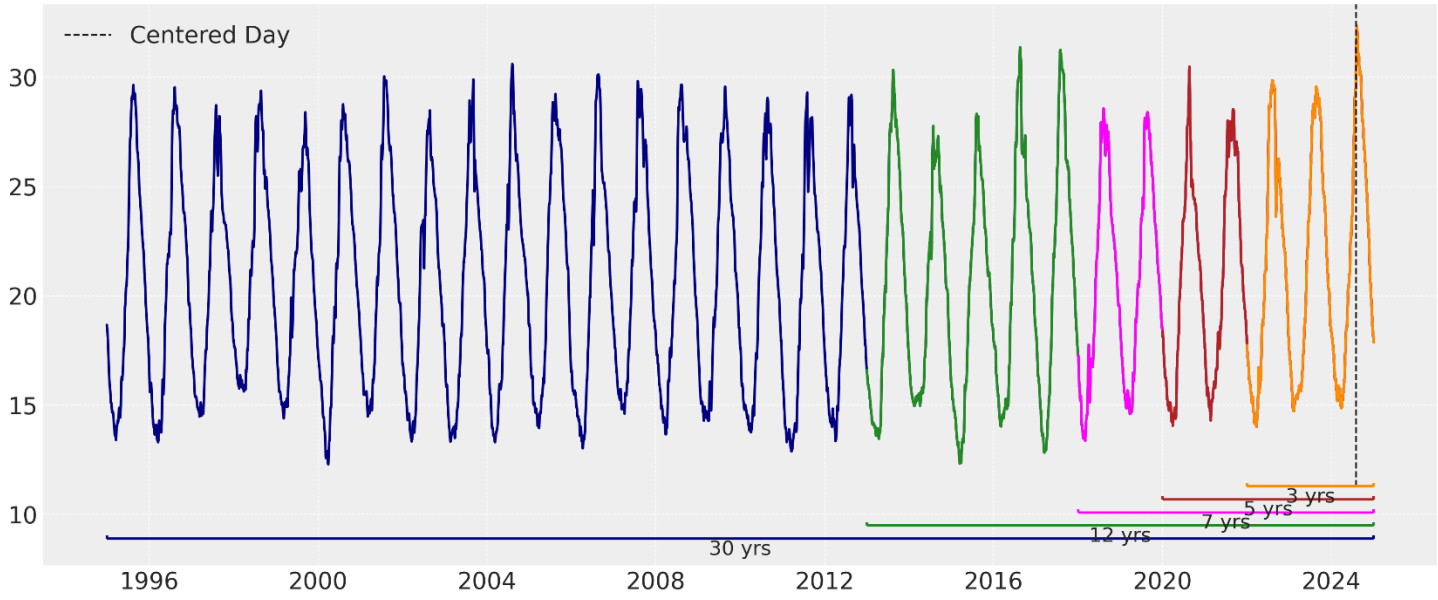


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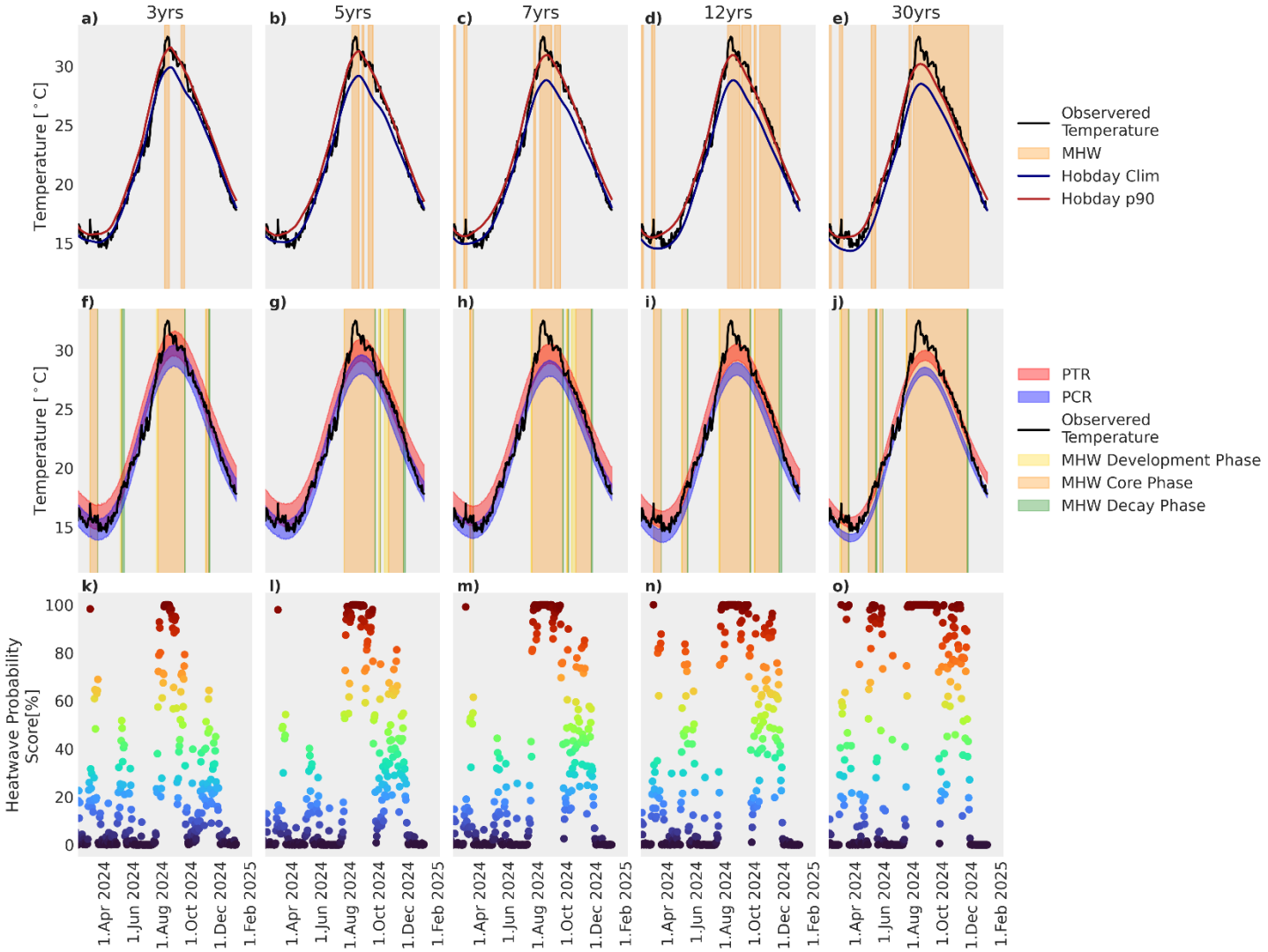


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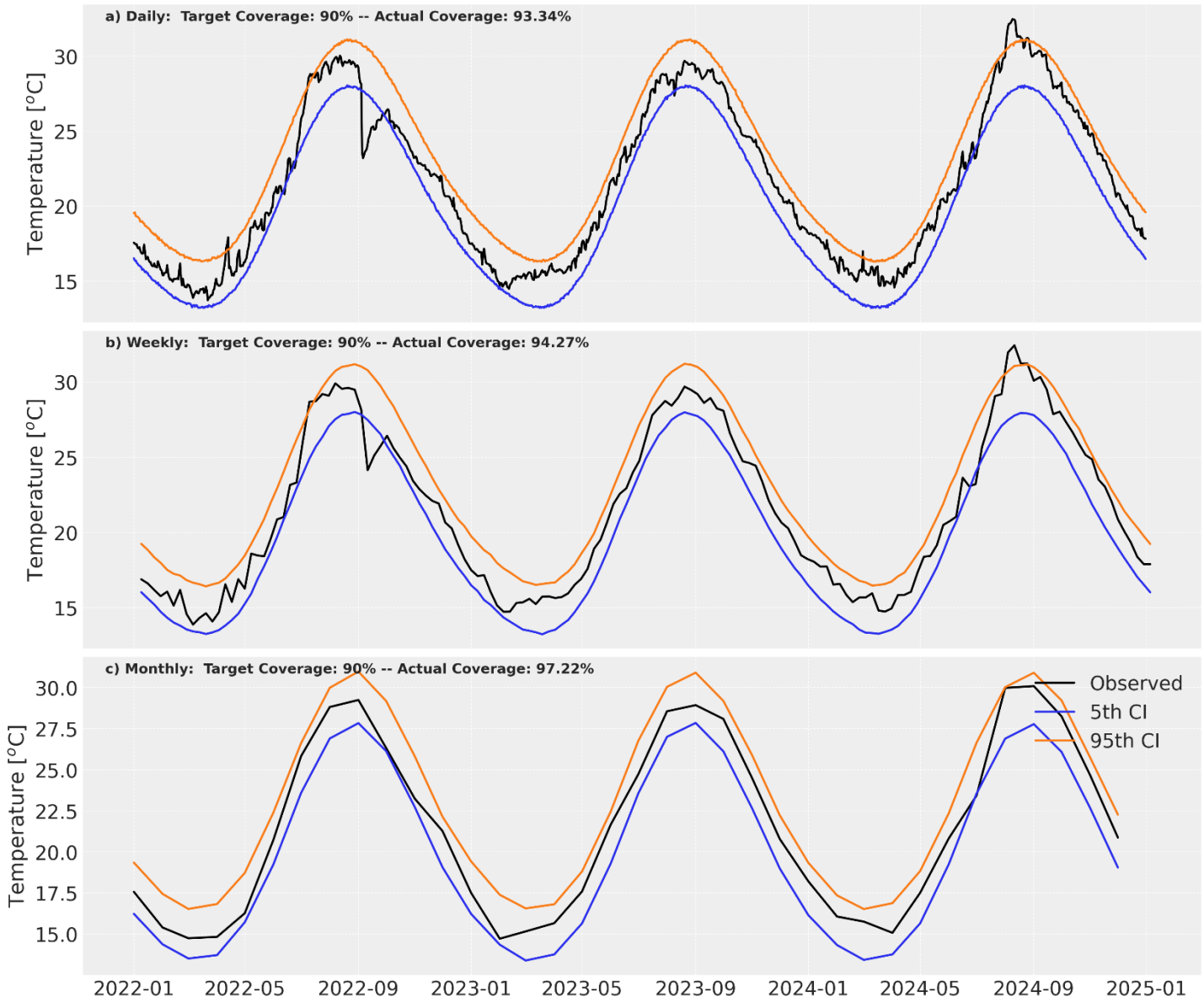


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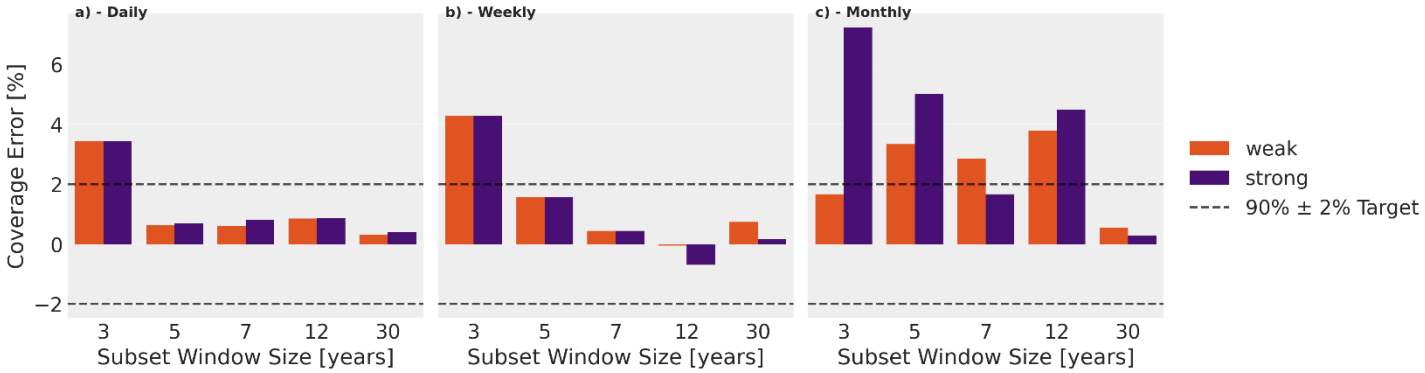


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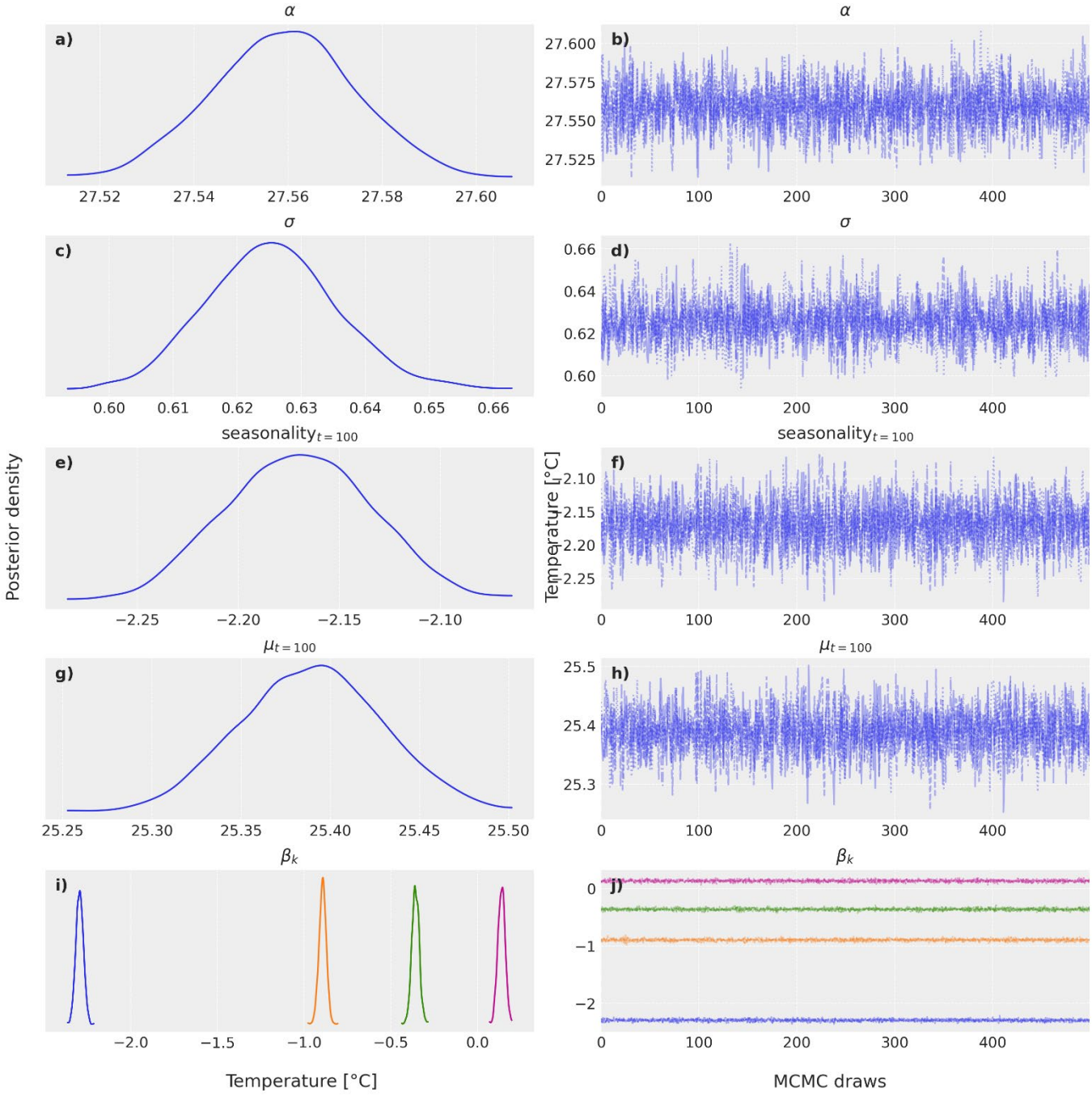


Figure S2.1. Same as Figure 4 but for site North West Shelf**.**

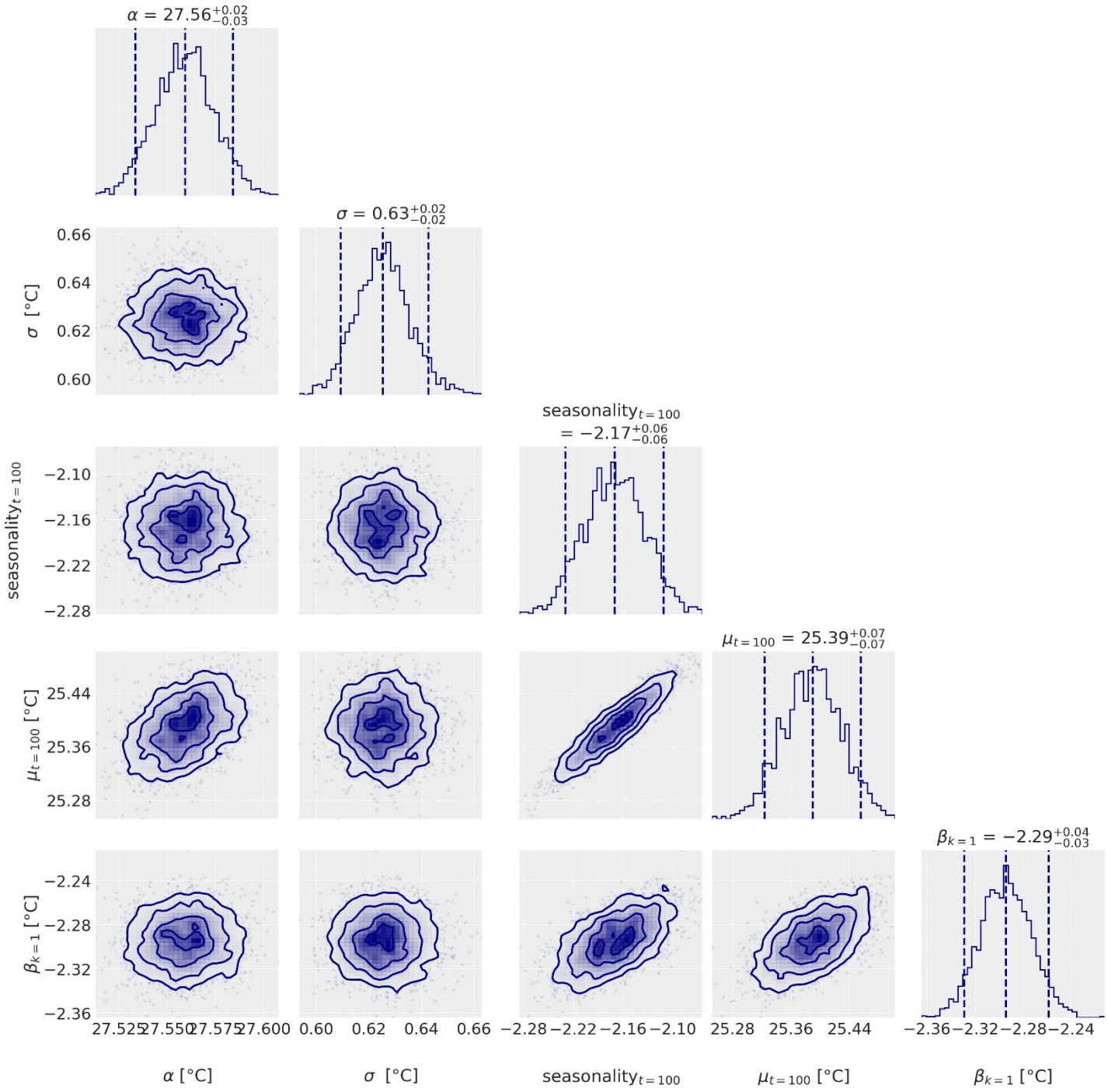


Figure S2.2. Same as Figure 5 but for site North West Shelf.

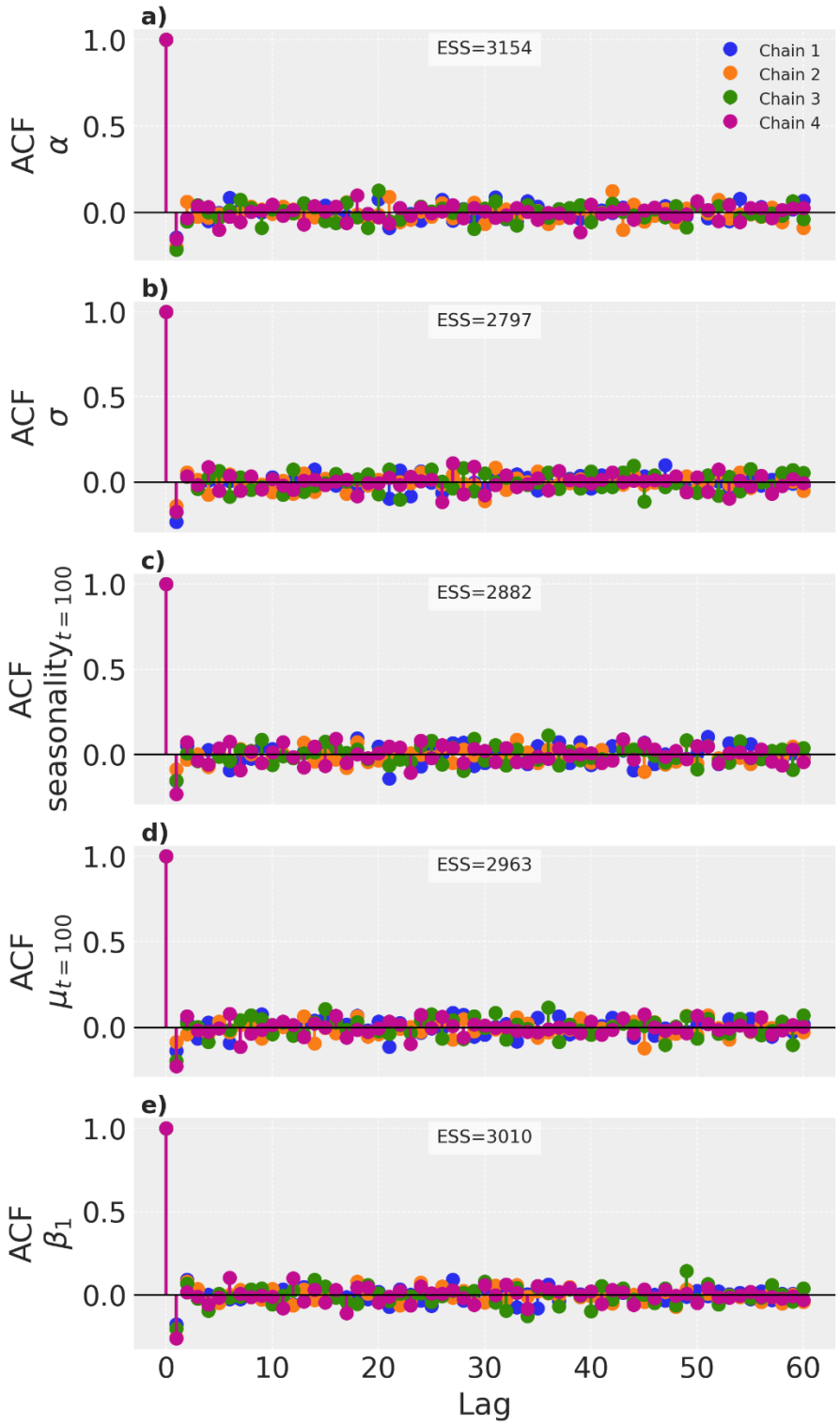


Figure S2.3. Same as Figure 6 but for site North West Shelf.

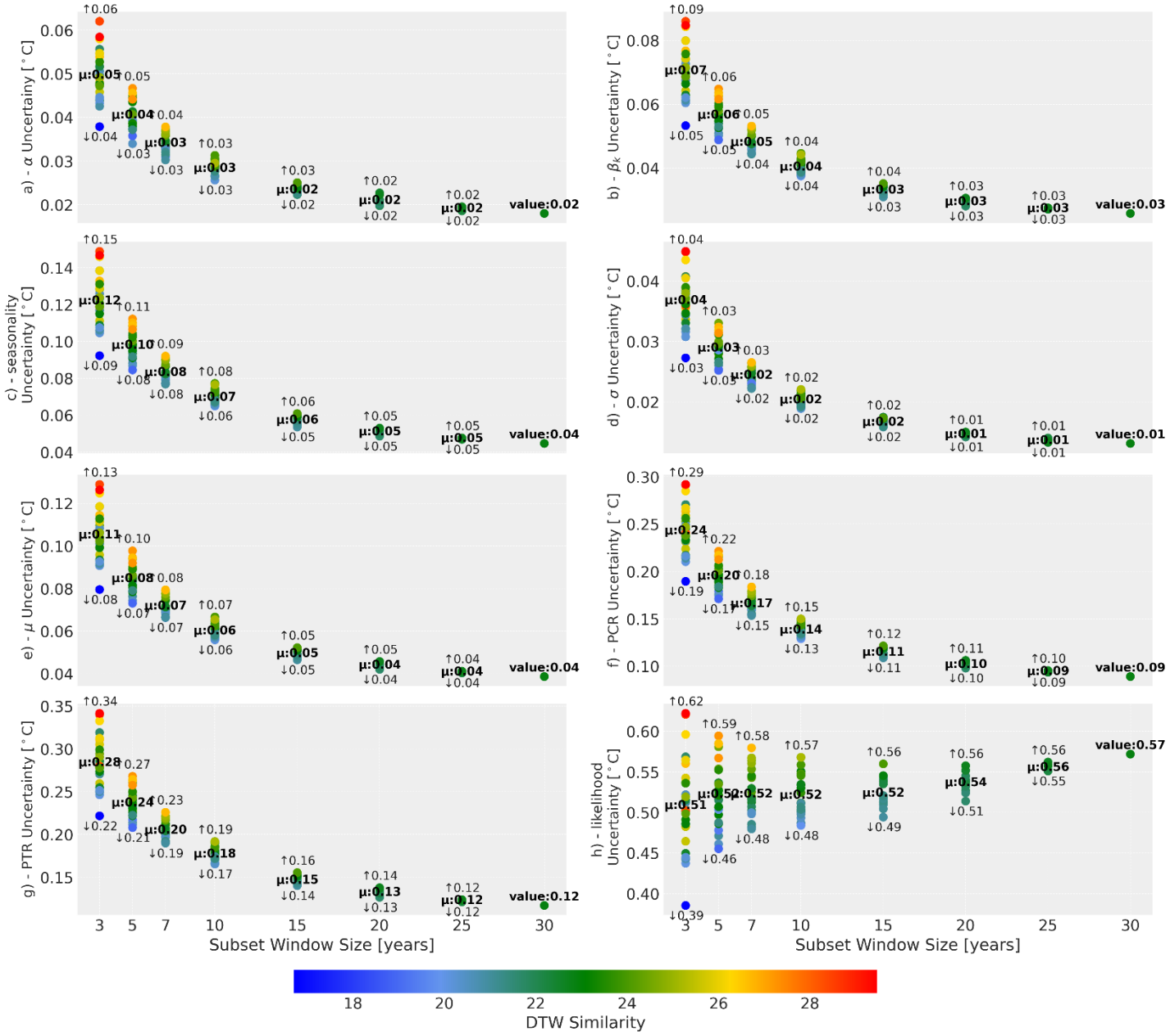


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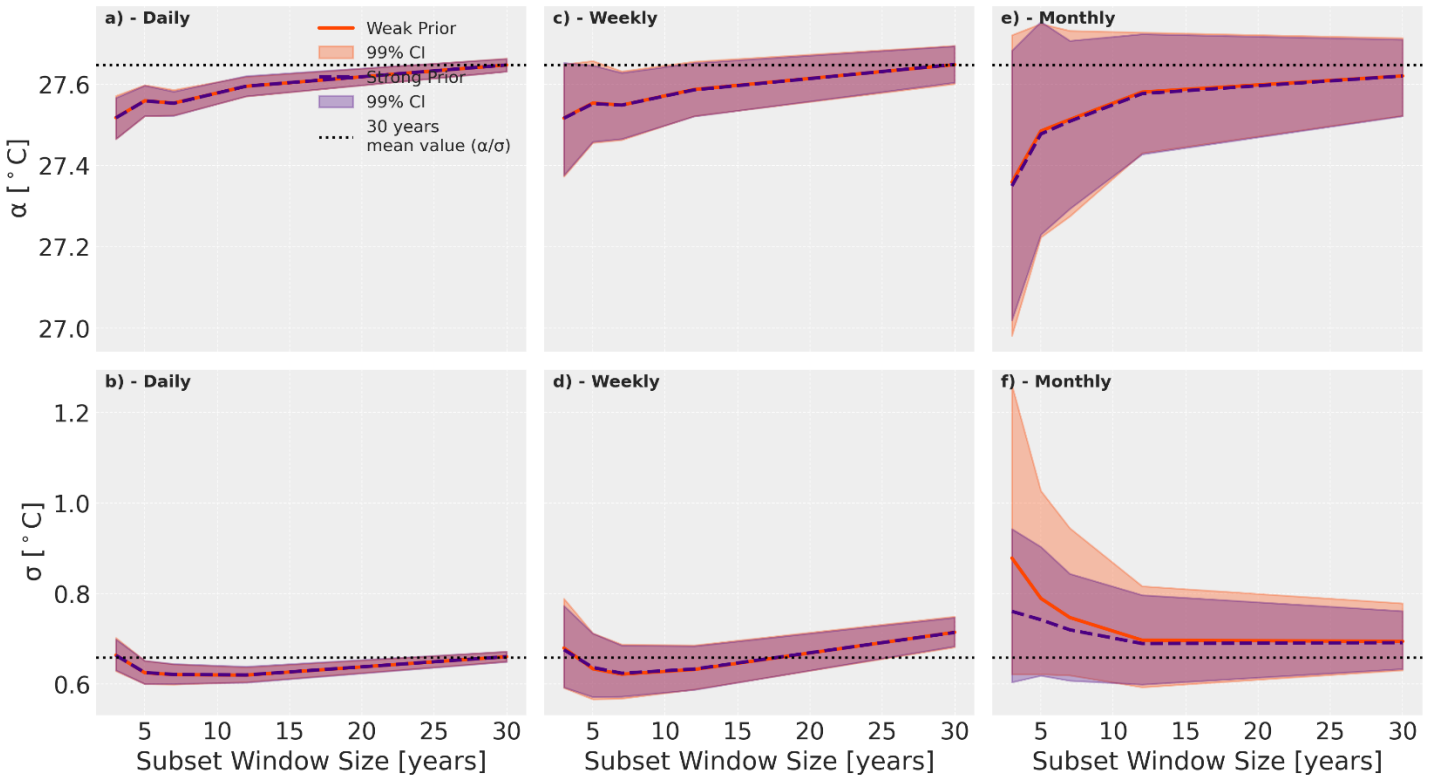


Figure S2.5. Same as Figure 8 but for site North West Shelf.

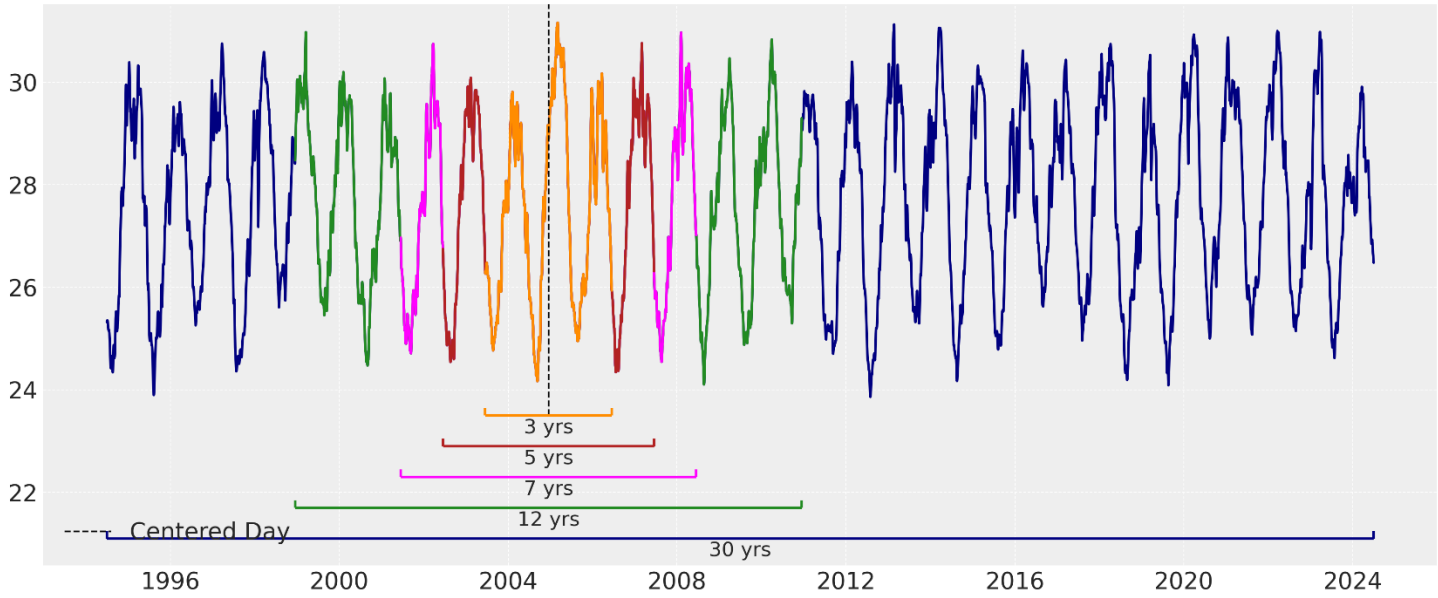


Figure S2.6. Same as Figure 9 but for site North West Shelf.

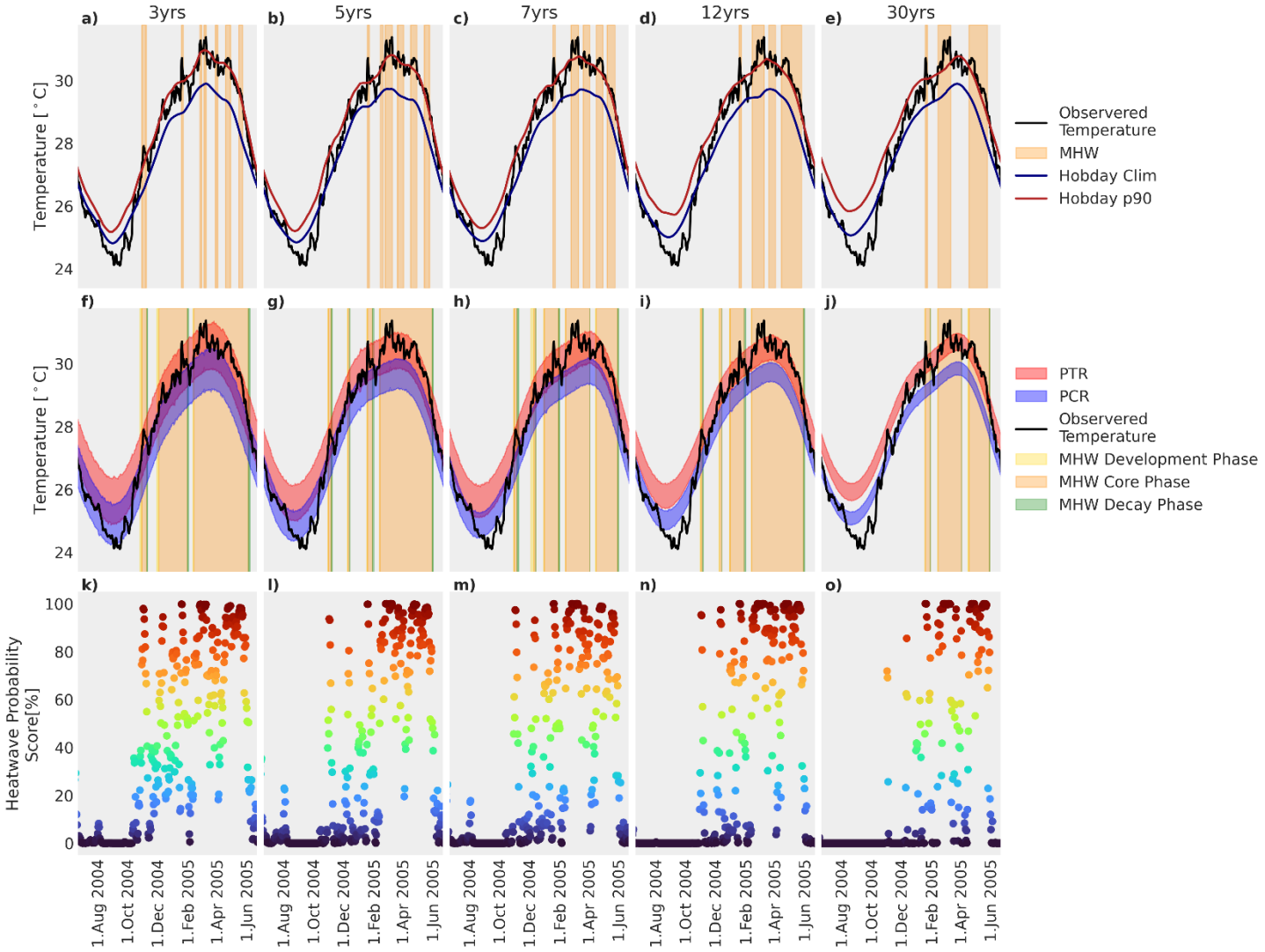


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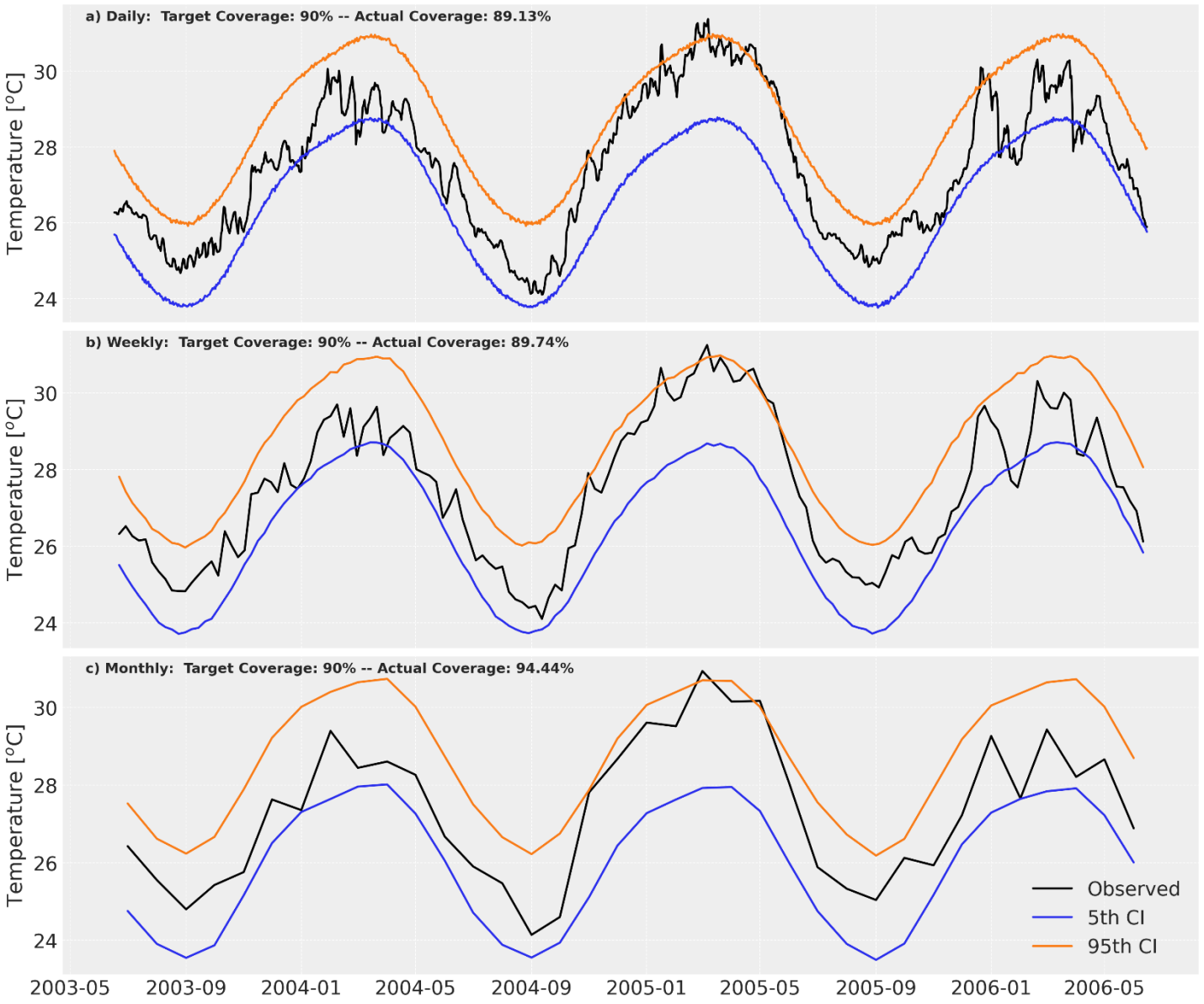


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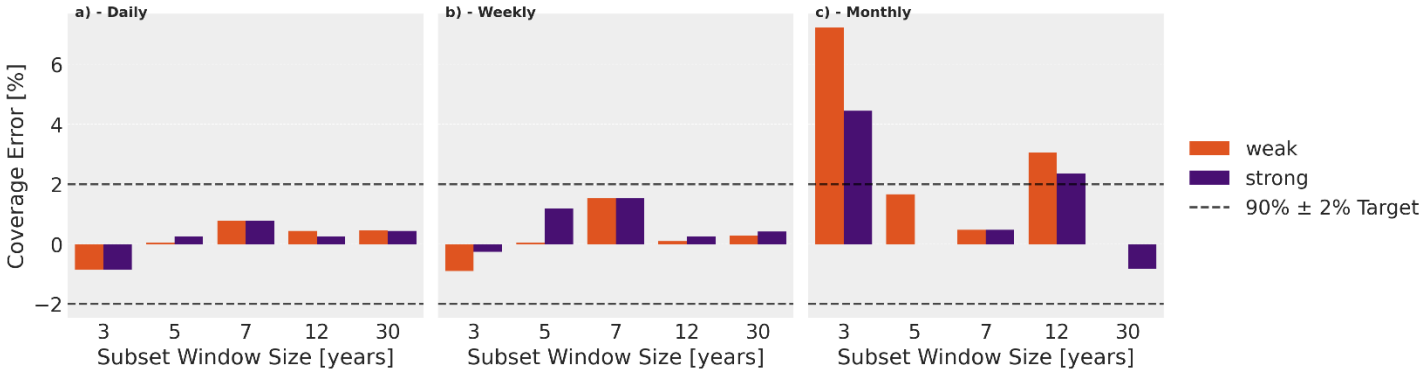


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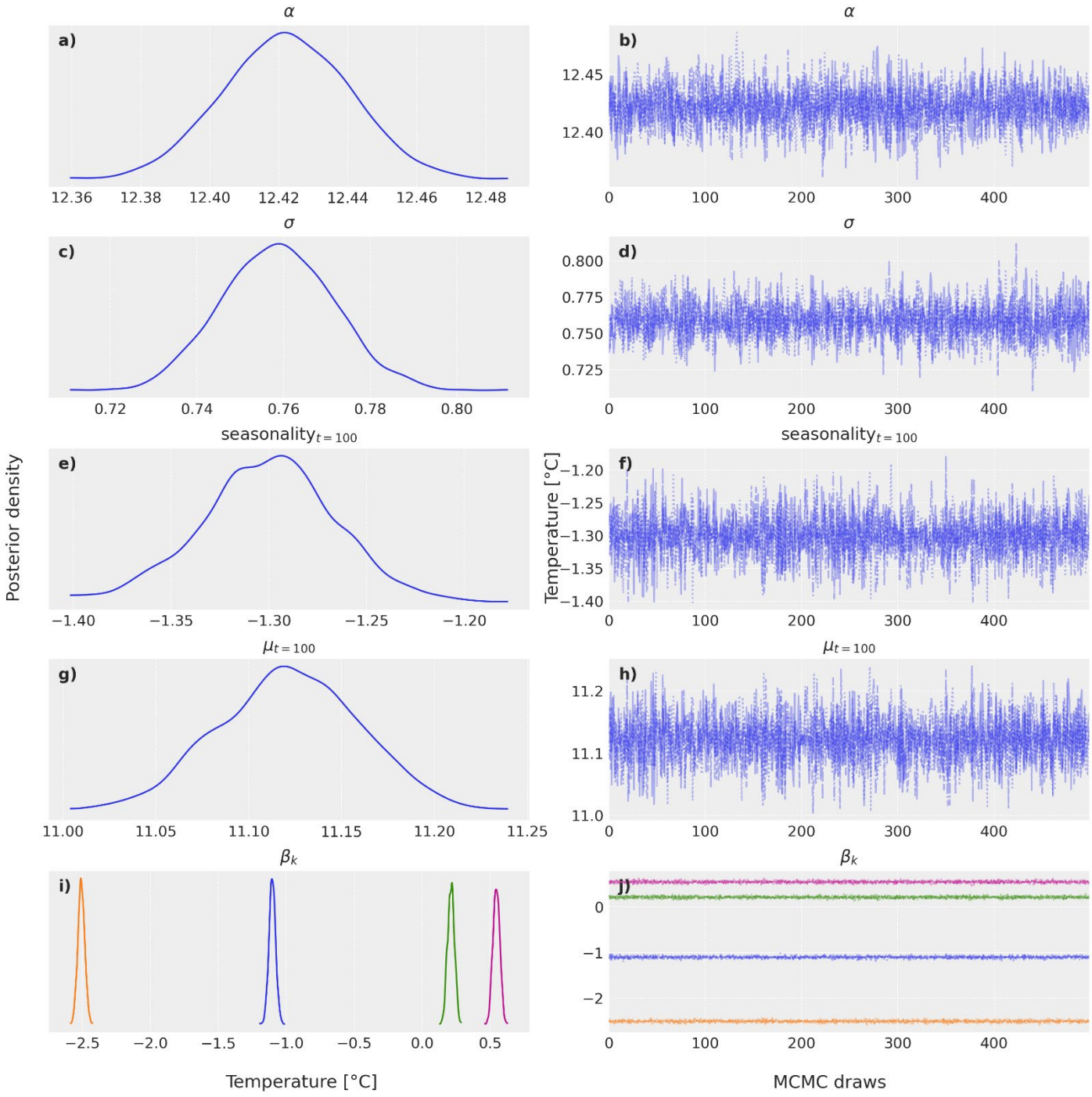


Figure S3.1. Same as Figure 4 but for site UK**.**

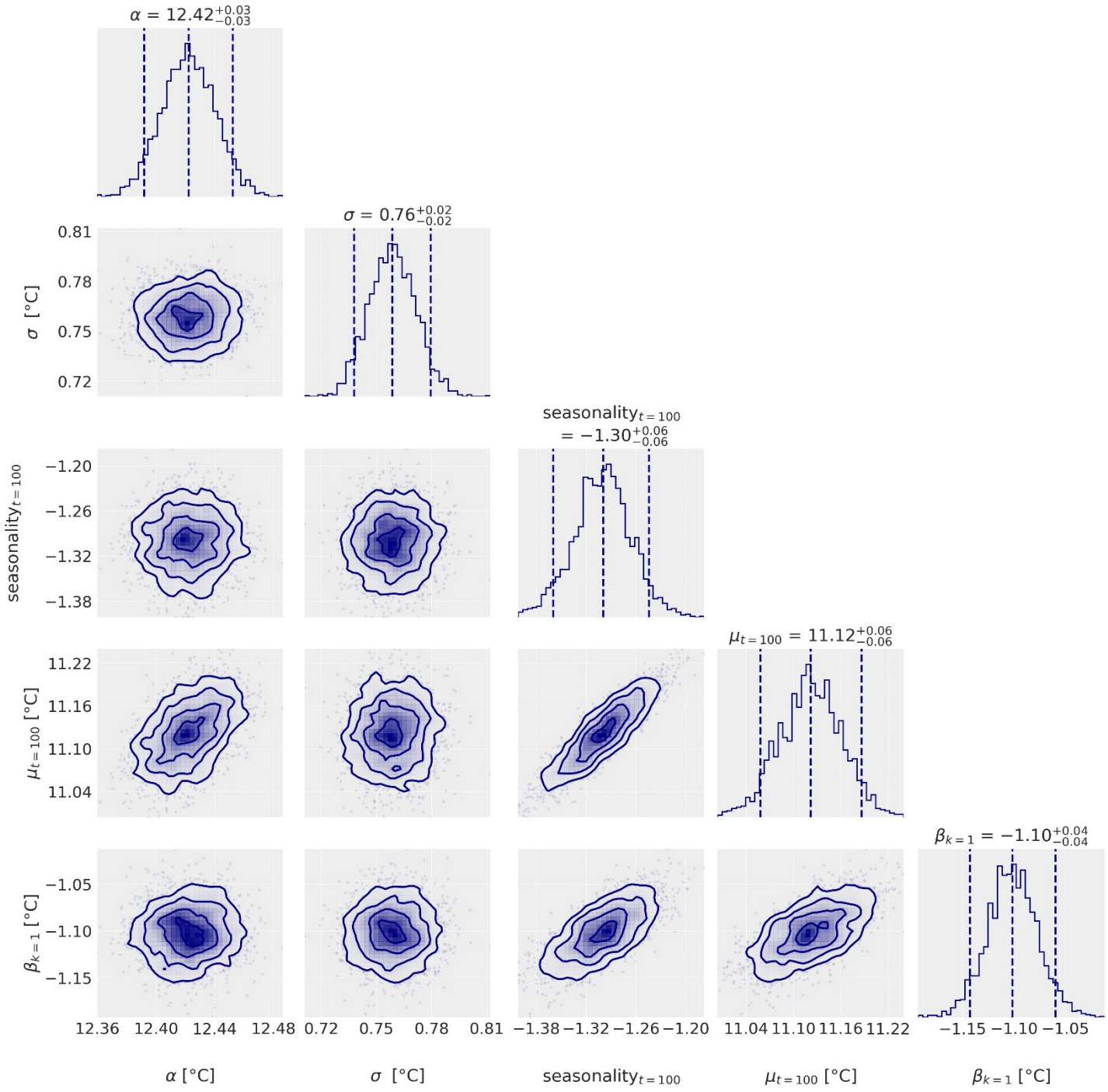


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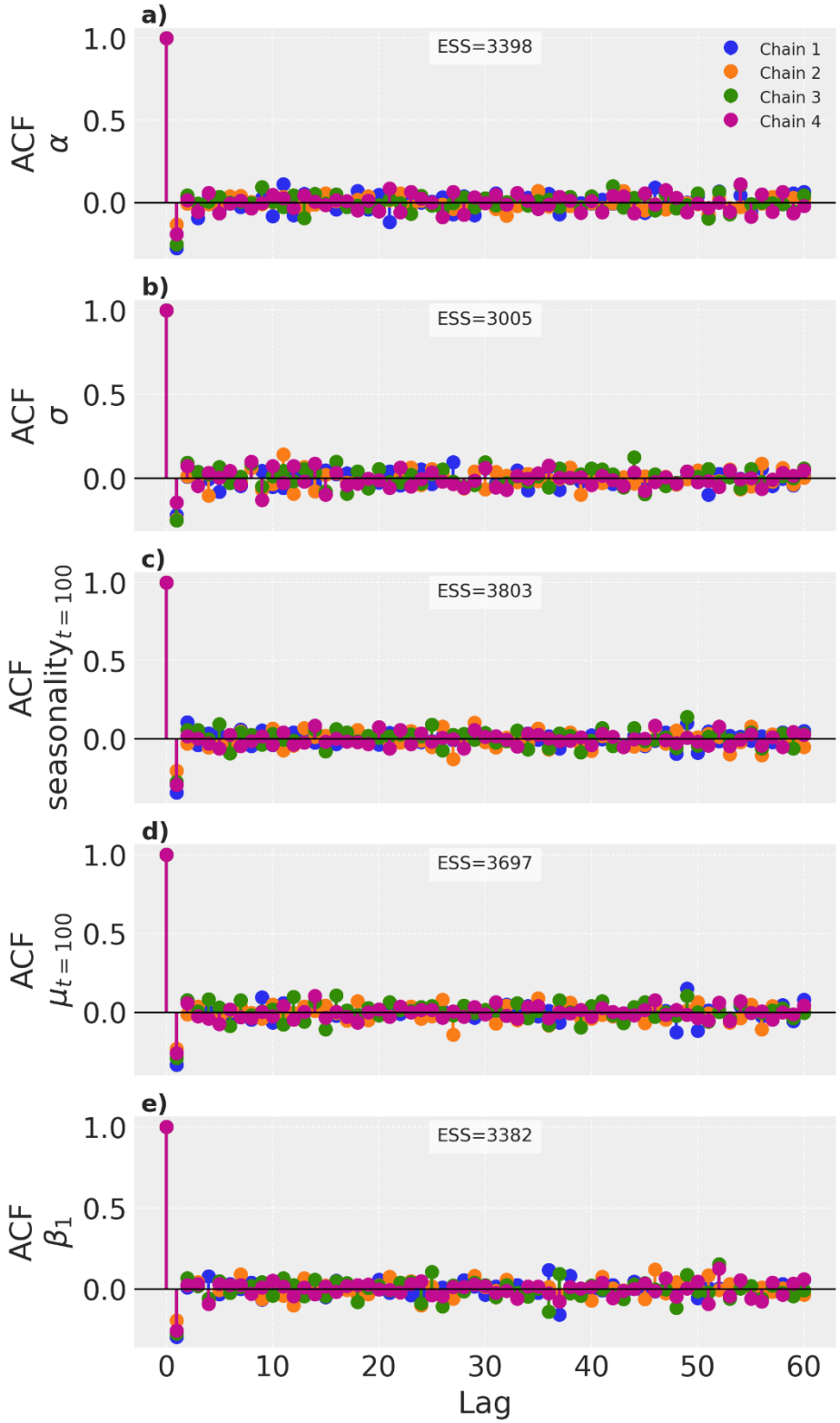


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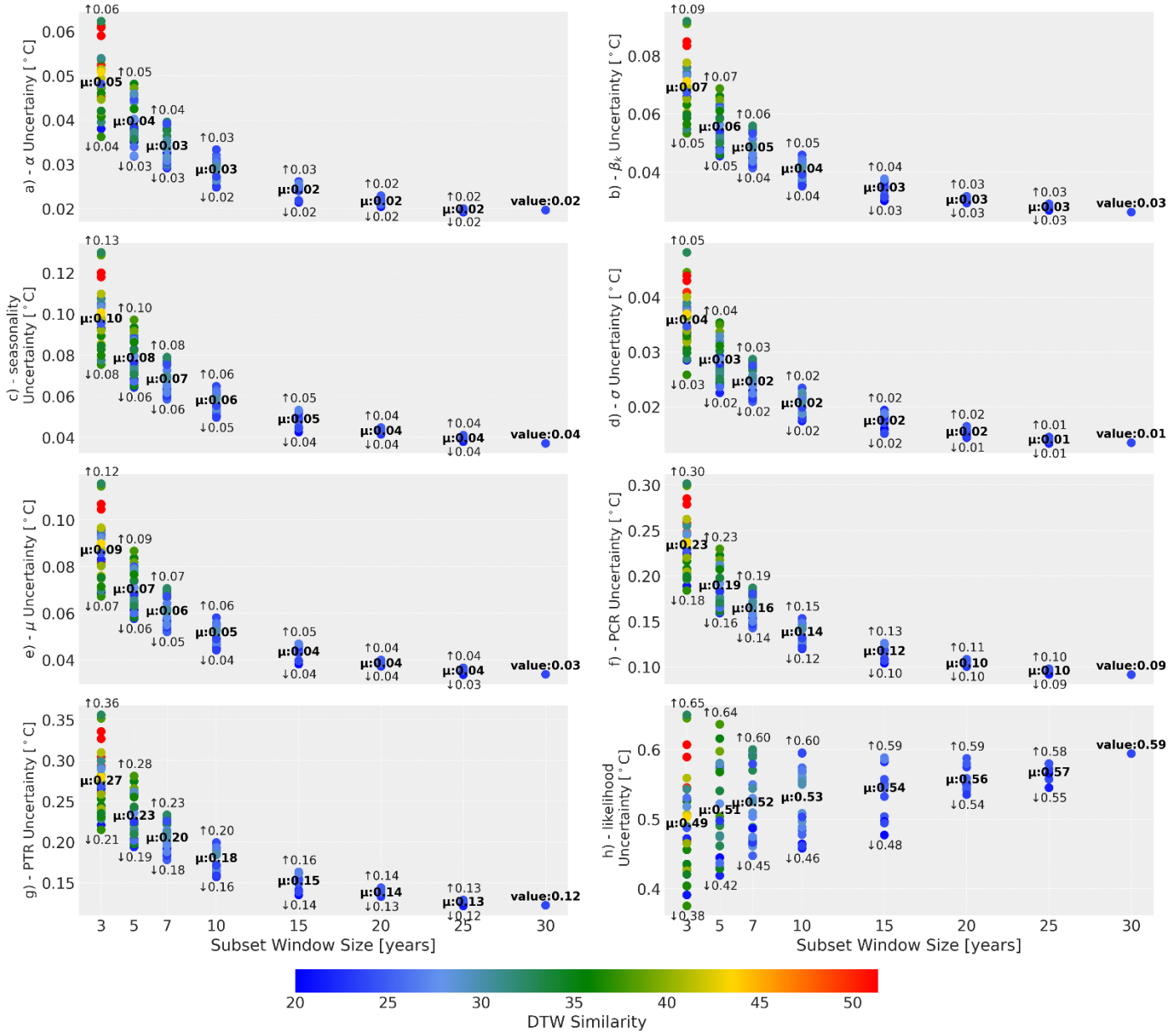


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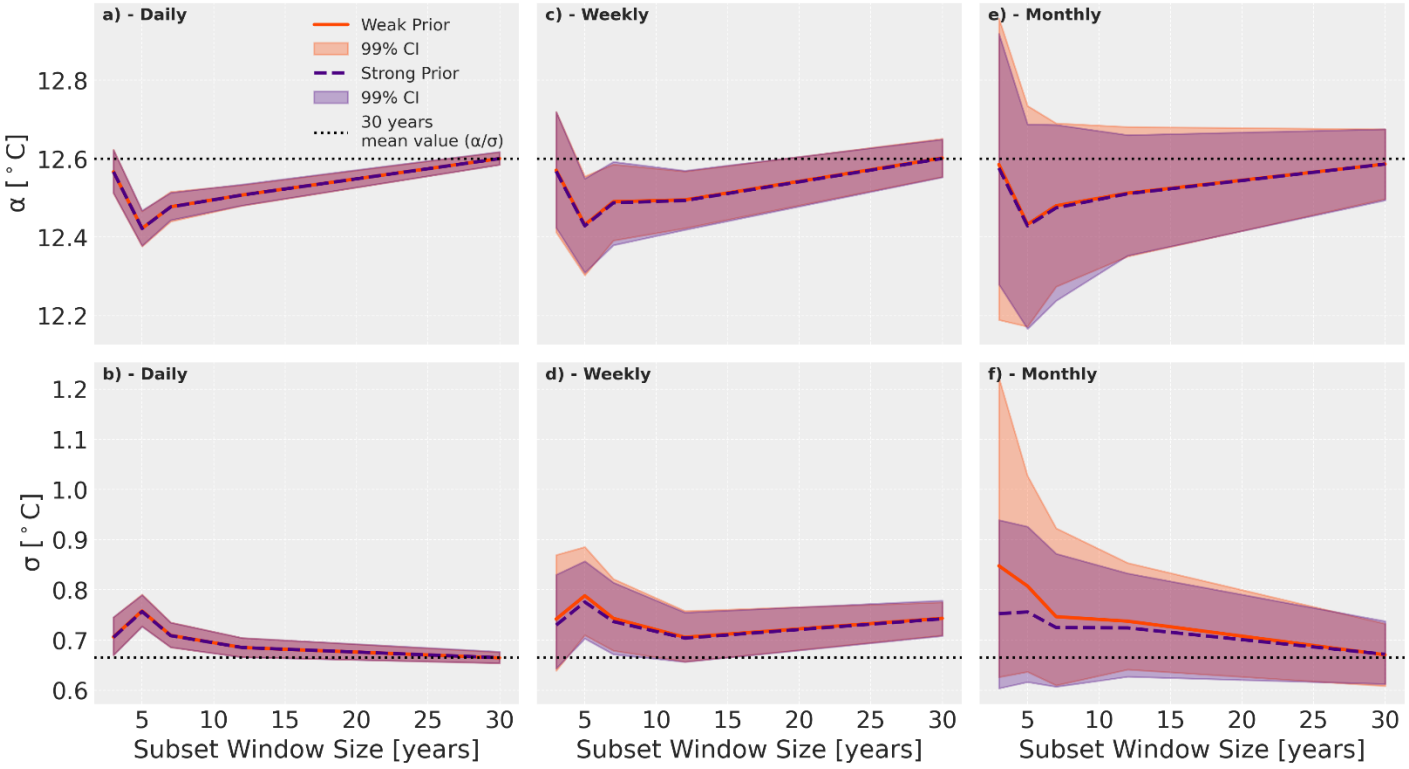


Figure S3.5. Same as Figure 8 but for site UK.

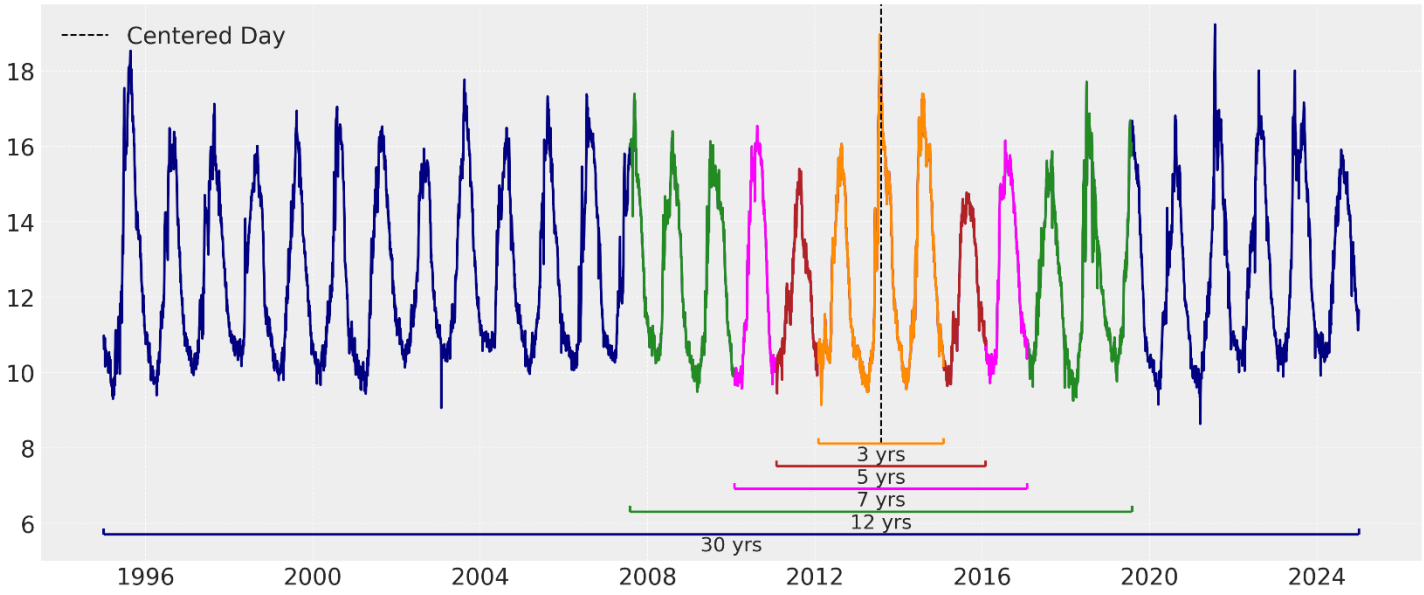


Figure S3.6. Same as Figure 9 but for site UK.

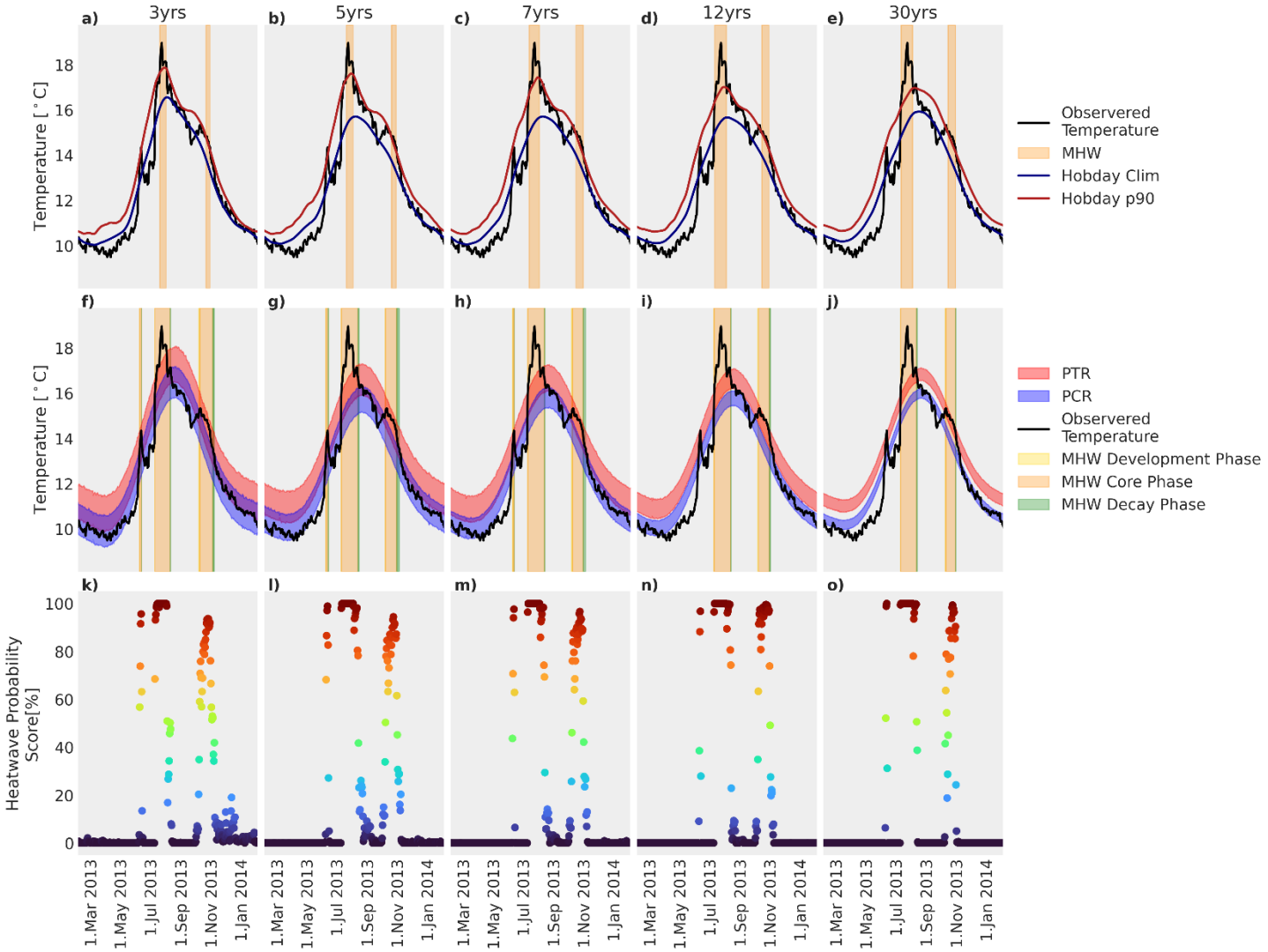


Figure S3.7. Same as Figure 10 but for site UK.

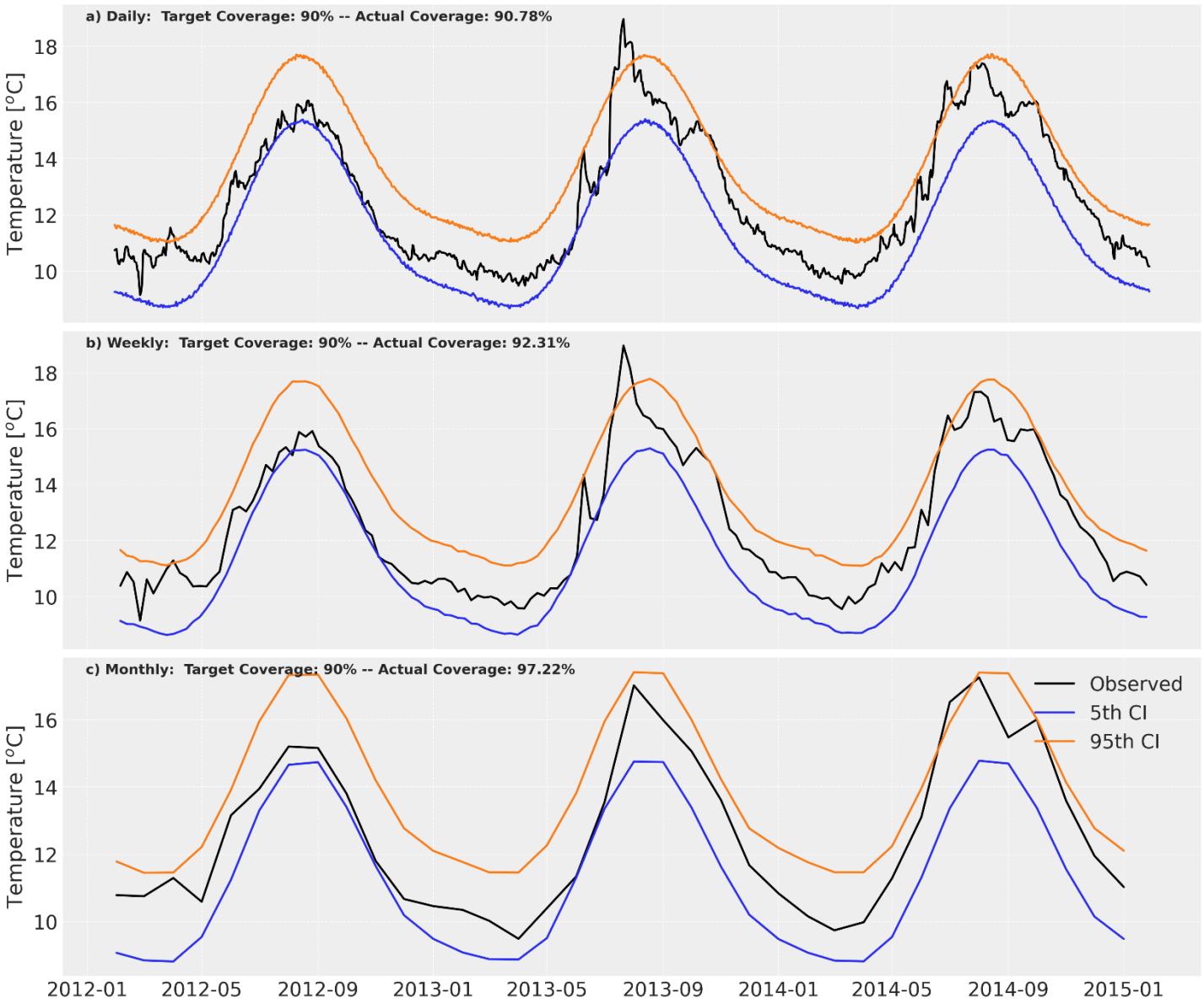


Figure S3.8 Same as Figure S0.1 but for site UK.

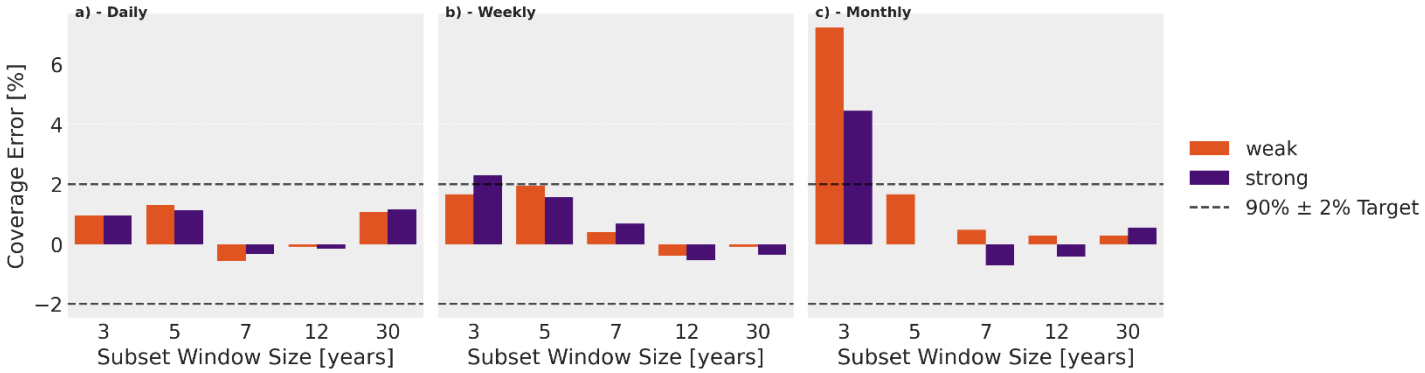


Figure S3.9 Same as Figure S0.2 but for site UK.